SRAVAN KUMAR PRAGADA

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EDUCATION

Northeastern University | *Boston, MA* Expected Dec 2026

Master of Science, Electrical and Computer Engineering (IoT) | ***GPA: 3.89/4.0***

Courses: Wireless Sensor Networks & IoT, Computer Hardware Security, Fundamentals of Computer Engineering, Solid State Devices.

Sagi Rama Krishnam Raju Engineering College | Andhra Pradesh, India April 2024

Bachelor of Technology, Electronics and Communication Engineering | ***GPA: 3.71/4.0***

Courses: Electronic Devices and Circuits, Digital Logic Designs, Embedded Systems, Microprocessors and Microcontrollers.

TECHNICAL SKILLS

Hardware & Tools: PCB Design, Soldering & Rework (SMD, Through-Hole), Wire Harness Assembly, Oscilloscope, DMM, Signal Generator, FLIR Thermal Camera, Electronic Load, Arduino, Raspberry Pi, ESP32, Circuit Analysis, Schematic Capture, Mixed-Signal Circuits, Power Management

Programming: Embedded C, C++, Python, Verilog, Shell/Bash

Software: Altium Designer, KiCad, LTspice, Quartus Prime, Shapr3D, Arduino IDE, Linux, GitHub, VS Code

Protocols & Standards: UART, SPI, I2C, MQTT, HTTP/REST, TCP/IP, UDP, Bluetooth/BLE, Wi-Fi, Zigbee, IPC/WHMA-A-620, ISO 13485

PROFESSIONAL EXPERIENCE

Electrical Engineer, Contractor| Glaukos Corporation | Massachusetts, USA Jun 2026 – Sep 2026

Electrical & Systems Engineer, co-op| Glaukos Corporation | Massachusetts, USA May 2025 – Jan 2026

- Designed 4 PCBs in Altium Designer (USB-C/USB-A, ESP32, Astable Multivibrator, Arduino Shield) per DFM rules, gaining hands-on proficiency in schematic, layout, and design rule compliance; fabricated 70 cables per IPC/WHMA-A-620 standards with ISO 13485 traceability documentation.
- Built and validated 6 next-gen ophthalmic device prototypes, verifying 35 test points per board using DMM, oscilloscope, waveform generator, variable power supply, and FLIR thermal camera to ensure electrical and thermal integrity across all assemblies.
- Identified and reported around 10 firmware command issues during prototype validation, systematically reproducing failure conditions and documenting expected vs. actual behavior to enable targeted fixes in subsequent firmware releases.
- Analyzed root-cause of thermal anomalies via FLIR camera - diagnosed swapped cables, poor solder joints, rotated inductor terminals, and incorrect footprints; reworked 6 PCB versions advancing all prototypes to next design phase.

Teaching Assistant - Digital Design | Northeastern University | Massachusetts, USA Jan 2026 – Apr 2026

- Delivered 7 lab assignments across 3 sections (over 100 students) using Intel DE1-SoC FPGA and Quartus Prime; validated in-person schematic implementations and circuit demonstrations, ensuring correctness against design specifications.
- Taught and validated students through combinational circuit design - binary adders, ripple carry and full adder implementations, 4-bit subtractor, 7- segment display decoder logic, and a fully functional Arithmetic Logic Unit (ALU) with multiple operation modes.
- Debugged and validated in sequential circuit design labs covering synchronous and asynchronous digital counters, FSM-based railroad crossing signal controller, and sequential logic design methodologies for real-world control applications.
- Mentored students through the most complex lab - a single-cycle RISC-V processor implementation covering instruction fetch, decode, and execute stages, register file design, and control unit logic on Intel DE1-SoC using Quartus Prime synthesis.

Student Ambassador – IoT & Hardware | AICTE Idea Lab | Andhra Pradesh, India Aug 2022 – Apr 2024

- Converted a physical lab room into a fully functional smart room using Arduino and C++, integrating motion, sound, touch, LDR, distance, water level, IR, DHT11, and ESP32 camera sensors to automate lighting control, water tap operation, outdoor light intensity regulation based on sunlight, and automatic tank refill via relay-controlled motor circuit.
- Implemented real-time IoT connectivity via MQTT protocol for web browser-based sensor monitoring and integrated ESP32 camera with Blynk app for live notifications; designed 12V to 5V/3.3V power supply PCBs in KiCad using through-hole components to power all IoT hardware modules across the Idea Lab.
- Led recurring technical workshops over 2 years for groups of 45 students per session covering IoT system design, PCB design fundamentals, embedded systems programming, and hands-on Arduino development for school and college students.

ACADEMIC PROJECTS

Digital Shopping Cart Dec 2023 – Apr 2024

- Built a smart shopping cart using Raspberry Pi (CPU), ESP32 (Wi-Fi), barcode camera scanner, and ADC-based weight sensors - architected for real-time product identification, in-cart weight tracking, and instant stock updates upon item placement.
- Validated system with 15 different items achieving barcode recognition up to 15cm from camera lens with stock updates within seconds of scanning; QR-based checkout directly launched the default payment app, eliminating traditional billing queues.
- Deployed in a real-world test environment and handed off to juniors for further development into handheld basket integration, demonstrating production-readiness and extensibility of the system architecture.

Smart Bike Keys Sep 2021 – Nov 2021

- Designed a keyless bike ignition system on Arduino Uno integrating FPM10A fingerprint sensor (162-print capacity) and RFID module - fingerprint authentication completes within 2 seconds of finger placement under normal conditions.
- Implemented dual-authentication logic with fingerprint as primary access method and RFID tag as fallback for adverse conditions (wet fingers, sensor damage), ensuring reliable ignition in all real-world scenarios; awarded 1st place at CRUX 2021.